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COMMON HALF YEARLY EXAMINATION - 2025

Standard XII

Reg.No. | | | | | | |

MATHEMATICS

Part - I

Time : 3.00 hrs *Marks : 90*

$20 \times 1 = 20$

I. Choose the correct answer:

1. If A^TA^{-1} is symmetric, then $A^2 =$
   a) A^T b) $(A^{-1})^2$ c) A^{-1} d) $(A^T)^2$
2. If $\rho(A) = \rho([A/B])$, then the system $AX = B$ of linear equations is
   a) consistent and has infinitely many solution b) inconsistent
   c) consistent and has a unique solution d) consistent
3. The value of $\sum_{n=1}^{13}\left(i^n + i^{n-1}\right)$ is
   a) 1 b) 0 c) $1+i$ d) i
4. If $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is
   a) 3 b) 4 c) 1 d) 2
5. A zero of $x^3 + 64$ is
   a) $4i$ b) -4 c) 0 d) 4
6. If α, β, γ and δ are the roots of the polynomial equation $2x^4 + 5x^3 - 7x^2 + 8 = 0$, then the value of $\Sigma\alpha\beta$ is
   a) $-7/2$ b) $5/2$ c) $7/2$ d) $-5/2$
7. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is
   a) [0,1] b) [−1,0] c) [1,2] d) [−1,1]
8. The amplitude for $y = \tan x$ is
   a) π b) 2π c) 1 d) cannot be defined
9. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11,2), the coordinates of the other end are
   a) (5,−2) b) (−2,5) c) (−5,2) d) (−3,2)
10. The angle between the line $\vec{r} = \left(\hat{i} + 2\hat{j} - 3\hat{k}\right) + t\left(2\hat{i} + \hat{j} - 2\hat{k}\right)$ and the plane $\vec{r}.\left(\hat{i} + \hat{j}\right) + 4 = 0$ is
    a) 45° b) 90° c) 0° d) 30°
11. The value of the limit $\lim_{x\to 0}\left(\cot x - \frac{1}{x}\right)$ is
    a) 2 b) ∞ c) 0 d) 1
12. The percentage error of fifth root of 31 is approximately how many times the percentage error in 31?
    a) 5 b) 31 c) $1/31$ d) $1/5$
13. Linear approximation for $g(x) = \cos x$ at $x = \pi/2$ is
    a) $x - \pi/2$ b) $-x - \pi/2$ c) $x + \pi/2$ d) $-x + \pi/2$

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14. The volume of solid of revolution of the region bounded by $y^2 = x(a-x)$ about x–axis is

a) $\frac{\pi a^3}{5}$ b) $\frac{\pi a^3}{6}$ c) πa^3 d) $\frac{\pi a^3}{4}$

15. $\int_{-\infty}^{\infty} \frac{1}{a^2+x^2}\,dx =$

a) $\frac{\pi}{2a}$ b) $\frac{\pi}{a}$ c) $\frac{\pi}{3a}$ d) $\frac{\pi}{4a}$

16. The solution of $\frac{dy}{dx} + P(x)y = 0$ is

a) $x = ce^{-\int Pdy}$ b) $x = ce^{\int Pdy}$ c) $y = ce^{\int Pdx}$ d) $y = ce^{-\int Pdx}$

17. A pair of dice numbered 1,2,3,4,5,6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is rolled and the sum is determined. Let the random variable X-denote this sum. Then the number of elements in the inverse image of 7 is

a) 3 b) 4 c) 1 d) 2

18. Let x have a Bernouli distribution with mean 0.4, then the variance of (2x–3) is

a) 0.6 b) 0.96 c) 0.24 d) 0.48

19. Which one of the following is a binary operation on N?

a) Division b) Subtraction c) Multeplication d) All the above

20. Which one of the following is incorrect?

a) $P \wedge P \equiv p$ b) $P \vee T \equiv P$ c) $P \vee F \equiv P$ d) $P \vee (P \wedge Q) \equiv P$

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. If $adj(A) = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$, find A^{-1}

22. Simplify $\left(\sin\frac{\pi}{6} + i\cos\frac{\pi}{6}\right)^{18}$

23. If $x^2 + 2(k+2)x + 9k = 0$ has equal roots, find k

24. Find $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

25. Determine whether the three vectors $2\hat{i} + 3\hat{j} + \hat{k}$, $\hat{i} - 2\hat{j} + 2\hat{k}$ and $3\hat{i} + \hat{j} + 3\hat{k}$ are coplanar.

26. Let $g(x) = x^2 + \sin x$. Calculate the differential dg.

27. Evaluate $\int_0^1 x^2(1-x)^3\,dx$

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28. The probability density function of X is given by $f(x)=\begin{cases} k\,xe^{-2x} & \text{for } x>0 \\ 0 & \text{for } x\le 0 \end{cases}$.

Find the value of k

29. Let $A=\begin{pmatrix}1&0&1&0\\0&1&0&1\\1&0&0&1\end{pmatrix}$, $B=\begin{pmatrix}0&1&0&1\\1&0&1&0\\1&0&0&1\end{pmatrix}$, $C=\begin{pmatrix}1&1&0&1\\0&1&1&0\\1&1&1&1\end{pmatrix}$ be any three Boolean matrices of the same type, find $(A \vee B) \wedge C$

30. Find the locus of the point of intersection of perpendicular tangents to the hyperbola $9x^2 - 16y^2 = 1$.

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) 7 x 3 = 21

31. Solve the following system of linear equations by Cramer's rule
$5x - 2y + 16 = 0$, $x + 3y - 7 = 0$

32. If $\omega \neq 1$ is a cube root of unity, show that $(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128$.

33. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$.

34. Find the centre and radius of the circle $3x^2 + (a+1)y^2 + 6x - 9y + a + 4 = 0$.

35. Find the equations of tangent and normal to the curve $y = x^2 + 3x - 2$ at the point (1,2)

36. Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}\,dx$

37. Find the differential equation corresponding to the family of curves represented by the equation $y = Ae^{8x} + Be^{-8x}$, where A and B are arbitrary constants.

38. If μ and σ^2 are the mean and variance of the discrete random variable X, and $E(x+3) = 10$ and $E(x+3)^2 = 116$, find μ and σ^2

39. Let * be defined on R by $(a * b) = a + b + ab - 7$. Is * binary on R? if so, find $3 * \left(\frac{-7}{15}\right)$.

40. If $f(x, y) = \frac{1}{\sqrt{x^2+y^2}}$, Prove that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = -f$

Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) Solve the following system of linear equations by Gaussian elimination method
$2x - 2y + 3z = 2$, $x + 2y - z = 3$, $3x - y + 2z = 1$

(OR)

b) Salt is poured from a conveyer belt at rate of 30 cubic meter per minute forming a conical pile with a circular base whose height and diameter of base are always equal. How fast is the height of the pile increasing when the pile is 10 meter high?

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42. a) Solve : $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

(OR)

b) Prove that $p\rightarrow(\neg q \vee r) \equiv \neg pv(\neg q \vee r)$ using truth table.

43. a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4 m when it is 6 m away from the point of projection. Finally it reaches the ground 12 m away from the starting point. Find the angle of projection

(OR)

b) Find the area of the region bounded by x–axis, the curve $y = |\cos x|$, the lines $x = 0$ and $x = \pi$.

44 a) Find the foci, vertices and length of major and minor axis of the conic $4x^2 + 36y^2 + 40x - 288y + 532 = 0$.

(OR)

b) The rate of increase in the number of bacteria in a certain bacteria culture is proportional to the number of present. Given that the number triples in 5 hours, find how many bacteria will be present after 10 hours.

45. a) Prove by vector method that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

(OR)

b) A rectangular page is to contain 24 cm^2 of print. The margins at the top and bottom of the page are 1.5 cm and the margins at other sides of the page is 1 cm. What should be the dimensions of the page so that the area of the paper used is minimum.

46. a) Find the parametric form of vector equation and Cartesian equations of the plane containing the line $\vec{r}=\left(\hat{i}-\hat{j}+3\hat{k}\right)+t\left(2\hat{i}-\hat{j}+4\hat{k}\right)$ and perpendicailar to the plane $\vec{r}.\left(\hat{i}+2\hat{j}+\hat{k}\right)=8$

(OR)

b) On the average, 20% of the products manufactured by ABC company are found to be defective. If we select 6 of these products at random and X denotes the number of defective products find the probability that i) two products are defective ii) atmost one product is defective iii) at least two products are defective

47. a) If $z = x + iy$ is a complex number, such that $\mathrm{Re}\left(\frac{z-1}{z+1}\right)=0$, find the locus of z.

(OR)

b) Solve $(x^3+2xy^2)dx + (y^3 + 2x^2y)dy = 0$

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